What does the app tell you?

IT 520 · October 19 · 20 minutes Name: _____

Work with a partner. Use words and set notation; explain your choices. Complete the exit question yourself.

> **Our demo:** each observation selects one label from **S = {Sunny, Rain, Snow, Fog}**. Rain and Snow indicate precipitation. Our fictional school starts **On-time** for Sunny/Rain and uses **Delay** for Snow/Fog.
> **Same predicates as the slides:** P(x) = precipitation reported; Q(x) = school delayed.
> **NOT P(x)** = no precipitation; **NOT Q(x)** = school on time.
> **Definitions:** a set has distinct members; |A| counts them. N ∩ O means both; N ∪ O means either or both. A proposition is true or false; a predicate becomes a proposition when its inputs are supplied. AND requires both; OR includes both. A function assigns each input exactly one output. Domain = allowed inputs; codomain = declared output set; range = outputs attained over the whole domain.

1. Select and count **(3 minutes)**

List the sets N = {x ∈ S : NOT P(x)} and O = {x ∈ S : NOT Q(x)}.

N = _____ O = _____

N ∩ O = _____ N ∪ O = _____ |N ∪ O| = ____

The log is [Rain, Fog, Rain]. Number of observations: ____ Number of distinct labels: ____

2. Test conditions **(4 minutes)**

Fill the table with T or F. Then give a counterexample to “Every report means school starts on time.”

Report x	NOT P(x)	NOT Q(x)	NOT P AND NOT Q	NOT P OR NOT Q
Sunny				
Rain				
Snow				
Fog				

Counterexample and reason: _____

Is P(Fog) a proposition even if it is false? Explain. _____

3. Read a mapping **(4 minutes)**

The start-time app uses T : S → {On-time, Delay}. Write its assignments:

Sunny → _____ Rain → _____ Snow → _____ Fog → _____

Domain: _____ Codomain: _____

Range: _____ Is T a function? Explain why shared outputs are allowed.

Extension if finished: J uses the same rule on domain {Snow, Fog}, with codomain {On-time, Delay}. What is J’s range? _____

4. Class feedback + individual exit **(after paired work)**

> **Full display dictionary:** 00 = Sunny · 01 = Rain · 10 = Snow · 11 = Fog.

The full display receives `10`. Its report is _____. The start-time app receives **Delay**.
List **all** possible original reports: _____. Can it tell whether precipitation was reported? Why?

Individual exit: Is a many-to-one mapping always a bug? Explain using the start-time app’s purpose.